Fundamentals of Chemistry

Science

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Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Fundamentals of Chemistry (A11330)

Short Title: Fundamentals of Chemistry
Faculty Science and Computing
Department Science
Cluster Chemistry
Author EOWENS
Credits 10 **Level:** Introductory

Description of Module / Aims

This is a fundamental chemistry course for non-chemistry specialists, and deals with basic concepts and techniques of chemistry relevant to pharmaceutical science. Topics include chemical and physical properties of matter, chemical bonding and molecular structure, industrial processes and their stoichiometry, volumetric analysis, pH and rates of reactions and chemical equilibrium. Laboratory exercises illustrate and emphasise important concepts encountered in the theory, and are of particular relevance to the pharmaceutical and health industries.

Programmes

CHEM-0018	Higher Certificate in Science in Good Manufacturing Practice and Technology (WD_SGMPT_C)	1 / 1 / M

Indicative Content

- Atomic structure, atomic bonding and intermolecular forces and their effect s on properties (chemical and physical) of chemical reagents.
- Types of chemical reactions including acid/ base and redox reactions and their stoichiometry.
- Chemical equilibrium.
- Reactions kinetics and factors that affect reaction rates.

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Distinguish between ionic/covalent bonding, outline inter-molecular bonding and relate it to physical and chemical properties of the system.
2. Distinguish between acid/base and redox reactions.
3. Explain the difference between acidity/alkalinity, calculate theoretical pH and calibrate a pH meter.
4. Describe chemical equilibrium including equlibrium constant and Le Chatelier's principle.
5. Describe the rate law of a simple reaction and discuss the factors that influence reaction rates.
6. Review typical chemical processes and their relevant stoichiometric calculations.
7. Communicate by written and verbal means the results of theory and practical-based exercises.
8. Demonstrate a keen awareness of the safe handling and disposal of chemicals and good laboratory practice (GLP).

Learning and Teaching Methods

- Lectures.
- Practicals.
- Tutorials.
- Independent learning.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	40%	1,2,3,4,5
Continuous Assessment	60%	
Practical	50%	7,8
Case Studies	10%	6

Assessment Criteria

<40%: No understanding of the basic fundamentals of chemistry. Inadequate practical skills in the associated laboratories.

40%-49%: Has a superficial knowledge of the fundamentals of chemistry. Has a basic level of competency and skills in the associated laboratories.

50%-59%: Shows a reasonable level of understanding of the fundamentals of chemistry. Displays good practical skills in the associated laboratories.

60%-69%: Shows a good knowledge of the subject area, as well as an ability to reason and analyse problems related to the area. Has a high level of skills in the associated laboratories.

70%-100%: All the above to an excellent level, in addition to excellent reasoning skills and focused well thought out arguments. Engages in debate in class and has a high level of efficiency in the associated laboratories.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture		24
Independent Learning		246

Essential Material

- "Fundamentals of Chemistry Moodle page." . http://www.moodle.ie

Supplementary Material

- Brown , T., H. Lemay, B. Burnsten and C. Murphy. _Chemistry: The Central Science_. 3rd Ed.. Australia: Pearson, 2013.

Requested Resources

- Room Type: Chemistry Lab